

Ayan Syed

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I'm an aspiring engineer driven by a passion for building robust, intelligent software at scale. I thrive on chasing big ideas with experience ranging from developing real-time ML systems, embedded firmware, and web apps for my non-profit, to interning at NASA and robotics startups. My background is anchored deep in the intersection of software and people, with direct experience in C++, Python, Linux, and debugging large-scale production software. As an Intern, I want to continue chasing my love for collaborating on big ideas and contributing to the next generation of software and technology.

Education

De Anza College

Computer Science - Class of 2028 - 4.0 GPA

- Data Structures & Algorithms, C++ & Python Series, Circuit Design, Object-Oriented Programming, Networking
- Pursuing transfer to UC Irvine, UCLA, or UC Berkeley for a B.S. in Electrical Engineering & Computer Sciences

Experiences

Cobot - "Collaborative Robotics"

Summer/Fall 2025 - **Software Engineering Intern**

- Lead daily hands-on software branch and release QA testing, while collaborating with 100+ software and hardware engineers to help maintain the codebase and handle bug fixes/debugging
- Working with C++ & Python using Linux, contributed to inter-robot networking and distributed computing systems over ROS2 TCP/IP Communication stacks and Docker images

NASA - AMES Research Center

Summer 2024 - **Data Research Intern**

- Preprocessed and analyzed 500GB+ of open source emission datasets, partnering with NASA Aviation Division data engineers, extracting market KPIs and growth areas using NumPy and Matplotlib in Python
- Worked with 3 PhD students to engineer a custom backend API using JAVA and SQLite, enabling users to query a sorted aviation emission database. Also, co-authored an alternative fuel technical paper; Published on N.T.R.S.

DA Engineering Club - AWS DeepRacer

President & Lead Model Developer

- Lead a top 100 Ranking team developing and training high-performance RL car-racing models on AWS Cloud Training

FTC Team 8872 Robopocalypse

2024 Team Captain & 2023 Technical Lead

- Captained team to the top 1% in NorCal Region and qualified for & competed at the 2024 World Championship

Projects

EEG Neurofeedback Powered Study Assistant

- Led 5-member independent research team that enhanced ADHD and student study focus rates by 23% by developing a real-time brain monitoring system utilizing OpenBCI, Fast Fourier Transform, and multi-threading
- Classified concentration levels and distraction events with 73% accuracy with multithreaded data filtering pipelines (Bandpass, Low/High Pass Filters) and training a SciKit-Learn Random Forest Classifier Machine Learning model
- Delivered personalized study insights and dynamic study pacing by integrating Gemini API on a web dashboard

Gemini AI Robotic Arm

- Using Python & C++, developed an Open-Source 5DOF 3D-printed Desktop Robotic Arm Platform. Utilized OpenCV to find, classify, and locate objects, then integrated Gemini and RAG for smart manipulation using voice

CometKitz

- Co-founded non-profit educating 200+ SCUSD elementary students by developing robotics kits 27% cheaper than the cheapest commercial option, using custom PCB, 3D-printed Chassis, and Arduino C++ firmware.
- Managed team of 20 and led classes introducing robotic software to grades 4-8 by building a full-stack Blockly web application that translated to sequential Arduino C++ code and complied with embedded firmware
- Secured over \$10K in R&D/execution funding by presenting and networking with Silicon Valley industry and partnering with local government to teach kits at 11 schools for free. Featured on NBC Bay Area

Skills

Languages/ Frameworks: Python, C++, Java, Assembly Language, Unity3D, JS, OpenCV, NumPy, SkLearn, APIs, ROS2, SQL, SQLite

Tools: Linux/Ubuntu, Github, Blender, Pandas, Matplotlib, JSON, LLMs, Embedded Systems, Arduino, AWS, RAG, CI/CD, Docker

Other: Networking, Product Design, TCP/ IP, KiCAD, Debugging. Reinforcement Learning, Databases, Cloud, Multithreading, OS